

Notes: Greenwood and Thesmar (JFE, 2011)

Stock Price Fragility

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1 Big picture

This paper studies how the ownership structure of a stock can create non-fundamental risk. In a frictionless representative-agent asset pricing model, current ownership should not matter: if some investors sell for liquidity reasons, other investors absorb the trade with no price effect. Greenwood and Thesmar instead start from the empirical premise that demand curves for individual securities are not perfectly flat. If owners are forced to trade for reasons unrelated to fundamentals, and if arbitrage capital is limited, then the stock price can move for non-fundamental reasons.

The core idea is **stock price fragility**: a stock is fragile when its current owners are likely to face large, correlated liquidity shocks. Fragility is high if ownership is concentrated among a few investors, if those investors have volatile flows, or if their flows are strongly correlated. The paper implements this idea using mutual fund holdings and mutual fund flows for U.S. stocks from 1990 to 2007.

Interpretation. The paper turns ownership composition into a measurable source of risk. A stock is not just risky because its cash flows are risky. It can also be risky because the people who hold it may need to trade at the same time.

2 Incremental contribution of the paper

Main incremental contribution. The paper builds an *ex ante*, ownership-based measure of non-fundamental price risk and shows that it predicts future volatility and comovement.

More specifically, the paper contributes in five ways.

1. **From realized flow pressure to *ex ante* fragility.** Earlier work studies realized mutual fund fire sales, fund flows, and price pressure. This paper asks whether the ownership structure today can forecast future non-fundamental risk.
2. **A general fragility formula.** Fragility combines ownership concentration with the variance-covariance matrix of investor liquidity shocks. This is richer than simply asking how much institutional ownership a firm has.
3. **A predictive test for stock volatility.** Using mutual fund ownership, fragility strongly predicts future total volatility and idiosyncratic volatility, even with controls for size, book-to-market, past returns, lagged volatility, turnover, and firm fixed effects.
4. **Extension to return comovement.** The paper defines *co-fragility*: two stocks comove if their owners face correlated trading needs. This provides an ownership-based explanation for excess comovement and factor betas.
5. **Interaction with arbitrageurs.** The paper studies whether hedge funds and active mutual fund trades dampen or amplify mutual-fund flow shocks. It shows that arbitrage is not always stabilizing.

Why this is new. The key shift is from looking only at firm fundamentals or realized trading to looking at the *network of owners*. The paper shows that the same stock can become more fragile when its owner base changes, even if the firm's fundamentals do not change.

3 Data and measurement

3.1 Data sources

The empirical analysis uses the following data sources:

- **Mutual fund holdings:** quarterly holdings from Thomson Financial, December 1989 through December 2007.
- **Mutual fund flows:** CRSP mutual fund data, where flows are changes in total net assets net of fund returns.
- **Stock returns and firm characteristics:** CRSP and Compustat.
- **Sample restriction:** U.S. common stocks with market capitalization greater than the NYSE median and with positive mutual fund ownership.

3.2 Mutual fund flows

For fund k , flows are calculated as

$$f_{kt} = TNA_{kt} - TNA_{k,t-1}(1 + R_{kt}),$$

where TNA is total net assets and R_{kt} is the fund return.

The paper estimates a variance-covariance matrix of percentage flows and then rescales by current fund assets to obtain the dollar flow covariance matrix Ω_t .

Interpretation. This is the empirical bridge between investor-level liquidity shocks and stock-level price risk. A stock held by funds with volatile and correlated flows is exposed to a larger demand shock.

4 Model and key empirical objects

4.1 Investor weights and flow-driven trading

Let w_{ikt} be the portfolio weight of stock i in investor k 's portfolio:

$$w_{ikt} = \frac{n_{ikt}P_{it}}{a_{kt}},$$

where n_{ikt} is shares held, P_{it} is price, and a_{kt} is investor portfolio value.

A change in investor demand can be decomposed into active rebalancing and flow-driven trading. The flow-driven component for stock i from investor k is approximately

$$w_{ikt}f_{kt}.$$

Aggregating across owners gives the stock-level flow-driven demand shock:

$$W'_{it}F_t,$$

where W_{it} is the vector of investor weights in stock i and F_t is the vector of investor flows.

4.2 Return equation

The paper assumes a stable relation between flow-driven demand and contemporaneous returns:

$$r_{i,t+1} = \alpha + \lambda \frac{W'_{it} F_t}{y_{it}} + \varepsilon_{i,t+1},$$

where y_{it} is stock market capitalization and λ measures the price impact of flow-driven trading.

Interpretation. If investors collectively sell a stock for liquidity reasons, returns fall if demand curves are downward-sloping. The larger λ is, the larger the price impact.

4.3 Fragility

Taking the conditional variance of the flow-driven demand term gives the paper's central measure:

$$G_{it} = \frac{1}{y_{it}^2} W'_{it} \Omega_t W_{it}.$$

Fragility is high when:

- ownership is concentrated;
- owners have volatile inflows and outflows;
- owners' flows are positively correlated;
- mutual fund ownership is large enough for these shocks to matter.

4.4 Co-fragility and fragility beta

The same logic extends to pairs of stocks. Co-fragility between stocks i and j is

$$G_{ijt} = \frac{W'_{it} \Omega_t W_{jt}}{y_{it} y_{jt}}.$$

Two stocks are co-fragile when their owners face correlated liquidity shocks. The paper also defines a fragility beta with respect to a portfolio p :

$$G_{it}^p = \frac{\text{cov}_t(W_{it} F_{t+1}, \sum_j s_{jt}^p W_{jt} F_{t+1})}{\text{var}_t(\sum_j s_{jt}^p W_{jt} F_{t+1})}.$$

Interpretation. A stock can have a high HML beta not only because its fundamentals are value-like, but also because it is held by funds whose flows move with value-stock fund flows.

5 Empirical specifications

5.1 First-stage validation

The first-stage test asks whether measured fragility predicts future flow-induced trading:

$$|W_{i,t+1} F_{t+1}| = a + b \sqrt{G_{it}} + Z_{it} \Gamma + u_{it}.$$

A positive b validates fragility as a forecast of future liquidity-driven demand volatility.

5.2 Volatility regressions

The main volatility regression is

$$\sigma_{i,t+1} = a + b\sqrt{G_{it}} + Z_{it}\Gamma + u_{i,t+1},$$

where $\sigma_{i,t+1}$ is next-quarter return volatility.

The paper estimates Fama-MacBeth regressions and also reports specifications with controls and firm fixed effects.

5.3 Comovement regressions

For stock pairs, the paper estimates covariance and correlation regressions:

$$\begin{aligned}\sigma_{ij,t+1} &= a + b\sqrt{G_{ijt}} + Z_{ijt}\Gamma + u_{ij,t+1}, \\ \rho_{ij,t+1} &= a + b\frac{G_{ijt}}{\sqrt{G_{it}G_{jt}}} + Z_{ijt}\Gamma + u_{ij,t+1}.\end{aligned}$$

Controls include industry similarity, size similarity, book-to-market similarity, and common owners.

5.4 Arbitrage interaction

To study liquidity provision, the paper estimates whether other traders' order imbalances move with mutual fund flow-driven trades:

$$D_{it}^X = \delta_{it} + \gamma_{it}D_{it}^{MF} + \nu_{it}.$$

Then it tests whether the volatility-fragility relation is stronger when other traders move in the same direction:

$$\sigma_{i,t+1} = a + b|1 + \hat{\gamma}_{it}| + c\sqrt{G_{it}} + d|1 + \hat{\gamma}_{it}|\sqrt{G_{it}} + u_{i,t+1}.$$

A positive d means that arbitrageurs or other traders amplify mutual fund liquidity shocks rather than absorb them.

6 Identification and empirical design

The paper is mainly a predictive design rather than a clean natural experiment. The key empirical question is whether ownership-implied liquidity-shock exposure predicts future volatility and comovement.

Strengths of the design.

- Fragility is computed using pre-existing ownership and historical flow covariance information.
- The paper controls for standard predictors of volatility such as size, book-to-market, past returns, turnover, lagged skewness, lagged volatility, and firm age.
- Fama-MacBeth estimation mitigates spurious common trends in volatility and mutual fund ownership.

- Firm fixed effects show that increases in a stock’s fragility are associated with increases in its volatility.
- Weighted regressions using mutual fund ownership give stronger estimates when mutual fund data better approximate total ownership.

Main limitations.

- Mutual funds are only one part of the ownership universe, so measured fragility is an imperfect proxy for total fragility.
- Ownership is potentially endogenous. Funds with volatile flows may choose stocks that are fundamentally volatile.
- Co-fragility results are especially vulnerable to omitted-variable concerns because funds with correlated flows may hold stocks with similar fundamentals.

Interpretation. The paper does not claim that all volatility is caused by mutual fund flows. It shows that an ownership-based exposure to liquidity shocks has strong predictive content for future volatility and comovement.

7 Main results

1. **Fragility predicts flow-induced trading.** Stocks with high fragility experience larger future absolute flow-induced trading.
2. **Fragility predicts future volatility.** The volatility-fragility relation is strong in raw sorts and remains significant with extensive controls.
3. **Mutual fund ownership alone is not enough.** What matters is the composition of ownership and the covariance structure of owner flows, not simply the fraction of shares held by mutual funds.
4. **Co-fragility predicts return comovement.** Stock pairs with high co-fragility have higher future covariance and correlation.
5. **Fragility betas help explain factor betas.** Ownership-flow structure explains a large share of HML and SMB comovement.
6. **Arbitrage is conditional.** Hedge funds and active mutual fund trades sometimes accommodate mutual fund liquidity shocks, but in some stocks they amplify them.

Economic message. Non-fundamental risk is partly an ownership-network phenomenon. Stocks become fragile when they are held by owners whose liquidity needs are large, volatile, and correlated.

8 Tables and figures

8.1 Figure 1 and Table 1: building the fragility measure

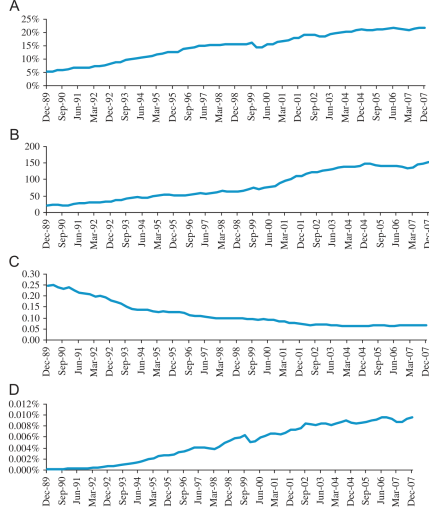


Fig. 1. Mutual fund ownership, concentration and fragility. For each characteristic, we plot the time-series of values for the median firm in each cross-section, drawn quarterly. The sample includes all stocks with market capitalization greater than the NYSE median between December 1989 and December 2007. Mutual fund ownership (Panel A) is the sum of shares owned by mutual funds, divided by shares outstanding. The number of mutual funds (Panel B) is the number of CRSP-listed funds owning a given stock. Ownership concentration (Panel C) is the sum of the squared shares held by mutual funds. Fragility (Panel D) is defined as the conditional expected variance of flow-driven net buys into a stock, and calculated according to Eq. (8) in the text. It is high when ownership is concentrated, or when mutual fund owners have volatile or correlated flows. Panel A: Mutual fund ownership as a fraction of shares outstanding. Panel B: The number of mutual fund owners. Panel C: Ownership concentration. Panel D: Fragility.

Mutual fund ownership, concentration, and fragility.

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- Figure 1 shows that mutual fund ownership rises sharply from 1989 to 2007, while ownership concentration falls.
- Despite lower concentration, measured fragility rises because mutual fund ownership and flow covariance become more important.
- Table 1 decomposes the inputs to the fragility measure: ownership concentration, number of owners, flow volatility, flow correlation, and the final fragility statistic.

Economic message: Fragility is not the same as ownership concentration. A stock can become fragile even when ownership becomes more dispersed, if owners' liquidity shocks are volatile and correlated.

Table 1
Constructing fragility.
The fragility of stock i in period t is given by

$$G_{it} = \frac{1}{\theta_i^2} W_{it} \Omega W_{it},$$

where W_{it} is the $K \times 1$ vector of mutual fund ownership of stock i ; Ω is the $K \times K$ variance-covariance matrix of fund flows; θ_i is stock i 's stock market capitalization. K denotes the number of funds. The table reports summary statistics for the components of fragility. H_{it} is the sum of the squared fund positions, scaled by total mutual fund ownership of stock i at date t (for instance, $H_{it} = 1$ if only one mutual fund owns the stock). The next rows show the number of mutual funds that hold a position in that stock for which we also have flow information. The bottom panels summarize the elements of the Ω matrix. For the standard deviation of fund flows σ_{it} , the unit of observation is at the fund k level, each quarter. For the correlation between flows $\rho_{kkt'}$ and its absolute value $|\rho_{kkt'}|$, the unit of observation is at the fund pair (k, k') level, each quarter. Summary statistics are shown in 1995Q4, 1999Q4, and 2003Q4. Variances and covariances in Ω are computed using all observations from 1989Q4 to t . Fragility is winsorized at 0.5% and 99.5% levels.

	Mean	Min	25%	Median	75%	Max
Ownership						
Concentration H_{it}						
Dec-1995	0.173	0.017	0.078	0.127	0.203	1.000
Dec-1999	0.132	0.015	0.059	0.092	0.153	1.000
Dec-2003	0.110	0.012	0.042	0.063	0.106	1.000
Number of Owners						
Dec-1995	78,716	1,000	31,000	53,000	102,000	571,000
Dec-1999	109,456	1,000	40,000	72,000	134,000	1001,000
Dec-2003	186,366	1,000	89,000	138,000	241,000	1278,000
Fund flows						
Flow volatility σ_{it}						
Dec-1995	0.103	0.000	0.042	0.094	0.158	0.337
Dec-1999	0.121	0.000	0.061	0.126	0.174	0.375
Dec-2003	0.140	0.000	0.088	0.145	0.186	0.375
Flow correlation $\rho_{kkt'}$						
Dec-1995	0.026	-0.990	-0.183	0.022	0.236	0.990
Dec-1999	0.037	-0.990	-0.130	0.026	0.205	0.990
Dec-2003	0.041	-0.990	-0.112	0.030	0.196	0.990
Flow correlation $ \rho_{kkt'} $						
Dec-1995	0.265	0.000	0.094	0.210	0.386	0.990
Dec-1999	0.229	0.000	0.071	0.168	0.329	0.990
Dec-2003	0.214	0.000	0.065	0.155	0.306	0.990
Fragility						
Fragility $G_{it} (\times 10e-4)$						
Dec-1995	0.651	0.000	0.075	0.270	0.761	6.865
Dec-1999	1.061	0.000	0.157	0.505	1.368	6.865
Dec-2003	1.274	0.000	0.324	0.813	1.811	6.875

Constructing fragility.

8.2 Table 2 and Figure 2: which stocks are fragile and how volatility behaves

Table 2

Characteristics of fragile stocks.

The sample includes all stocks that are owned by one or more mutual funds and which have end-of-quarter market capitalization above the NYSE median. The sample period is from December 1989 to December 2007. The table shows summary statistics for fragility-sorted portfolios, where fragility is the conditional expected variance of flow-driven net buys and computed according to Eq. (8) in the text. Stocks are sorted into portfolios based on end-of-quarter fragility. Active weight is the sum of the changes in weights adjusted for portfolio growth and adjusted for stock price appreciation; active weights are aggregated across funds to the individual stock level. BE/ME denotes the book-to-market ratio. The bottom rows of the table report statistics on the number of stocks in the portfolio at different points in time.

	Fragility Quintile				
	Low	2nd Quintile	Middle	4th Quintile	High
Fragility G %	0.210	0.460	0.675	0.959	1.562
Fragility $G(t-1)$ %	0.227	0.471	0.681	0.951	1.496
N Owners	92	168	150	132	118
MF Ownership %	5.331	12.349	16.767	21.074	28.657
Active weight %	0.026	0.028	0.064	0.148	0.445
NYSE decile	7.915	8.234	7.960	7.630	7.228
BE/ME	0.512	0.515	0.521	0.483	0.405
MOM decile	5.320	5.376	5.483	5.628	5.675
Returns (%)					
Past quarter	3.325	3.431	3.672	4.446	4.394
Past 2 quarters	6.406	6.687	7.537	9.101	9.628
T (in quarters)	73	73	73	73	73
N (Dec-1989)	120	120	120	120	119
N (Dec-1995)	217	217	217	217	216
N (Dec-2000)	281	281	280	281	280
N (Dec-2005)	256	256	256	256	255
N (All)	16,423	16,386	16,387	16,387	16,360

Characteristics of fragile stocks.

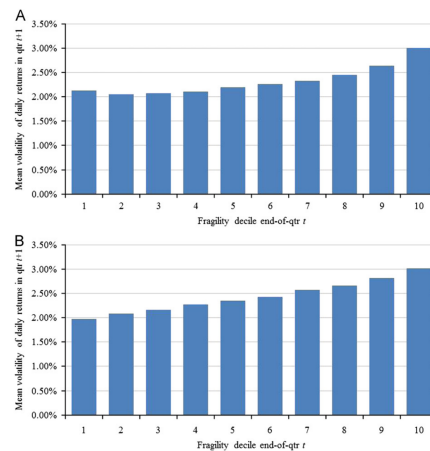


Fig. 2. Fragility and volatility. Each quarter, stocks are sorted into deciles according to their fragility. For each decile of fragility, we first compute, each quarter between 1989Q4 and 2007Q3, the mean volatility (standard deviation) of daily stock returns in the next quarter. The figures show the time series averages of these means, by decile of fragility. In Panel A, we draw on the full sample. In Panel B, we restrict the sample to firms with at least 20% of mutual fund ownership. Panel A: Full sample. Panel B: Stocks whose mutual fund ownership exceeds 20% of shares outstanding.

Fragility and volatility.

Read:

- Table 2 sorts stocks by fragility. More fragile stocks have higher mutual fund ownership, stronger past returns, lower book-to-market ratios, and more active mutual fund buying.
- Figure 2 shows that future return volatility rises with fragility. The pattern is stronger and more linear among stocks with higher mutual fund ownership.

Economic message: The raw data already show the central prediction: stocks with more fragile ownership structures have greater future volatility.

8.3 Table 3 and Figure 3: volatility and co-fragility

Table 3

Fragility and stock return volatility.

The dependent variable is the one-quarter-ahead standard deviation of daily stock returns or excess returns in that quarter:

$$\sigma_{it+1} = \alpha + \beta_1 \sqrt{G_{it}} + \beta_2 C_{it} + u_{it+1}$$

G_{it} denotes fragility, and Z_{it} is a list of control variables. The sample period is 1990–2007. Except for columns 3 and 7, we report Fama-MacBeth estimates, which are equal-weighted quarter-by-quarter, and t -statistics in brackets. In column 3, observations are weighted by mutual fund ownership at the end of the quarter. Column 7 shows a panel regression which includes a stock-level fixed effect, with standard errors clustered by period. The regressions in columns 5–7 further include a set of unreported controls: the log of unadjusted stock price, the log of market capitalization, the ratio of book equity to market equity, the past 12 month stock return, lagged skewness of stock returns, the log of age and share turnover. The remaining independent variables, whose coefficients are reported in the table, are: fragility, fragility calculated as if liquidity shocks of the different owners were uncorrelated (i.e., setting all off-diagonal terms of G_{it} to 0 in Eq. (8)), fragility calculated using only the correlation terms of liquidity stocks (i.e., setting all on-diagonal terms of G_{it} to 0 in Eq. (8)), the log of the number of mutual funds owning the stock, the fraction of stocks held by mutual funds, the ownership Herfindahl index H , the log of firm size, and the quarter- t volatility of returns. In the last three columns, the dependent variable is the standard deviation of excess returns, where excess returns are alternately based on the single-factor market model, the three-factor Fama and French (1993) model, or the three-factor Fama and French model plus momentum. FM denotes Fama-MacBeth estimation.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	Total return volatility σ							Excess return volatility:			
								1-Factor σ	3-Factor σ	4-Factor σ	
$\sqrt{G}$		0.696	0.711		0.793	0.226	0.699	0.152	0.583	0.564	0.556
		[15.17]	[16.67]		[12.66]	[6.27]	[7.45]	[4.81]	[16.97]	[17.27]	[17.29]
$\sqrt{G}$ (Diag)				0.489							
				[11.75]							
$\sqrt{G}$ (Off-diagonal)				0.524							
				[9.46]							
Log (# owners)	−0.001										
	[−6.57]										
MF share	0.022				−0.006	0.000	−0.030	0.000			
	[18.12]				[−3.89]	[0.03]	[−5.57]	[0.16]			
Ownership Herfindahl H					−0.002	−0.001	−0.004	−0.002			
					[−2.86]	[−1.16]	[−1.41]	[−1.88]			
$\sigma(t)$								0.524			
								[36.38]			
Constant	0.024	0.018	0.018	0.018	0.019	0.029	−0.020	0.014	0.016	0.016	0.016
	[15.80]	[23.68]	[26.74]	[23.88]	[21.94]	[20.07]	[−1.21]	[9.05]	24.07	24.89	24.86
Observations	81,962	81,962	81,962	81,962	81,962	48,906	48,906	48,906	75,495	75,495	75,495
R-squared	0.06	0.08	0.11	0.09	0.09	0.48	0.57	0.58	0.08	0.08	0.08
Additional controls	No	No	No	No	No	Yes	Yes	Yes	No	No	No
Estimation	FM	FM	FM	FM	FM	FM	Panel + FE	FM	FM	FM	FM
Weighting	EW	EW	MF-weight	EW	EW	EW	EW	EW	EW	EW	EW

Fragility and stock return volatility.

Co-fragility and comovement of stock returns.

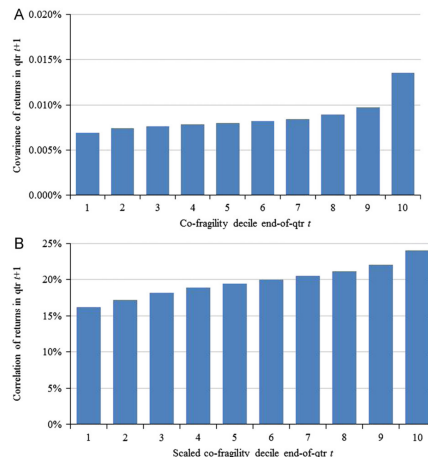


Fig. 3. Co-fragility and comovement of stock returns. Co-fragility is defined in Eq. (18) and measures the covariance of mutual fund flow-driven net buys into any pair of stocks i and j . Each stock pair in the sample is sorted into co-fragility deciles. For each decile of fragility, we compute, each quarter between 1989Q4 and 2007Q3, the mean covariance (Panel A), and the mean correlation (Panel B) of daily stock returns in the next quarter. The figures show the time-series average of these means. Panel A: Covariance. Panel B: Correlation.

Read:

- Table 3 reports formal volatility regressions. The coefficient on $\sqrt{G}$ is positive and highly significant across specifications.
- The effect survives controls for mutual fund ownership, ownership Herfindahl, firm characteristics, firm fixed effects, and lagged volatility.
- Figure 3 shows that stock-pair covariance and correlation rise across co-fragility deciles.

Economic message: Fragility is not just a visual sorting pattern. It has strong predictive power for return volatility, and its pairwise version predicts stock return comovement.

8.4 Table 4 and Table 5: comovement regressions and return horizon

Table 4

Co-fragility, covariance, and correlation.

In panel A, the dependent variable is alternately the one-quarter-ahead covariance between daily returns of stocks i and j :

$$\sigma_{ij,t+1} = a + b\sqrt{G_{it}G_{jt}} + Z_{it}C + u_{ij,t+1},$$

where G_{it} denotes co-fragility of stocks i and j , and Z is a list of pair-level control variables. In Panel B, the dependent variable is the one-quarter-ahead correlation between daily returns of stocks i and j :

$$\rho_{ij,t+1} = a + b \frac{G_{ij}}{\sqrt{G_{it}G_{jt}}} + Z_{it}C + u_{ij,t+1}.$$

In Panel B, we rescale co-fragility G_{ij} between stocks i and j by the square root of the product of their individual fragilities G_{it} and G_{jt} . The control variables include dummy variables for whether stocks i and j are in the same two-digit, three-digit, or four-digit SIC code, the absolute difference in log size between i and j , the absolute difference in log BE/ME between i and j , and one-quarter lagged correlation and covariance of stock returns between i and j . Regressions are estimated quarterly between December 1989 and December 2007 and include all stock pairs drawn from the largest 500 stocks each quarter. The table reports average regression coefficients and the associated Fama and MacBeth t -statistics.

	Panel A: Dependent variable = $\sigma_{ij,t+1}$				Panel B: Dependent variable = $\rho_{ij,t+1}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
G_{ij}		1.690 [4.06]	1.497 [4.10]	1.135 [3.94]				
$G_{ijt}/G_{it}G_{jt}$					0.088 [17.38]	0.057 [8.32]	0.047 [7.48]	
$SIC_{2it} = SIC_{2jt}$	0.344 [4.94]		0.329 [4.90]	0.243 [5.68]	0.057 [9.65]	0.055 [9.70]	0.048 [9.39]	
$SIC_{3it} = SIC_{3jt}$	0.386 [3.20]		0.348 [3.06]	0.272 [3.62]	0.045 [4.88]	0.047 [5.29]	0.039 [4.66]	
$SIC_{4it} = SIC_{4jt}$	0.113 [1.45]		0.141 [1.88]	0.125 [2.20]	0.055 [7.90]	0.053 [7.87]	0.048 [7.33]	
Common owners (log)	0.027 [0.95]		0.012 [0.39]	0.012 [0.47]	0.028 [11.50]	0.021 [10.58]	0.018 [9.45]	
Similar size	-0.011 [-1.21]		-0.014 [-1.53]	-0.010 [-1.60]	-0.003 [-3.32]	-0.004 [-4.24]	-0.003 [-3.61]	
Similar BE/ME	-0.021 [-5.44]		-0.016 [-4.64]	-0.013 [-4.41]	-0.004 [-10.07]	-0.003 [-8.39]	-0.003 [-7.85]	
σ_{ij}				0.295 [13.36]				
ρ_{ij}							0.194 [17.22]	
Observations	2,916,545	3,911,280	2,916,545	2,796,062	2,916,545	3,907,003	2,913,266	2,792,768
R-squared	0.04	0.04	0.08	0.17	0.06	0.02	0.07	0.11

Co-fragility, covariance, and correlation.

Table 5

Forecasting volatility and comovement at longer horizons.

Univariate regressions forecasting the standard deviation of returns in the next quarter, the covariance of returns in the next quarter, or the correlation of returns in the next quarter, based on daily, weekly, bi-weekly, and monthly returns. G_{it} denotes fragility of stock i and G_{jt} denotes the cross-fragility between stocks i and j . The first column of each panel repeats the results from Tables 3 and 4. The remaining columns re-estimate the regressions, with volatility and comovements based on weekly, bi-weekly, and monthly sampling of returns. In Panels A and B, the regression coefficients are adjusted so that long-horizon results are directly comparable with the results from the daily regressions. Regressions are estimated quarterly between December 1989 and December 2007; the table reports average coefficients from the 73 quarterly regressions and the associated Fama and MacBeth t -statistics. The constant term is omitted.

Measurement interval	Panel A: Total return volatility = $\sigma_{it,t+1}$				Panel B: Covariance = $\sigma_{ij,t+1}$				Panel C: Correlation = $\rho_{ij,t+1}$			
	Daily (1)	Weekly (2)	Bi-weekly (3)	Monthly (4)	Daily (5)	Weekly (6)	Bi-weekly (7)	Monthly (8)	Daily (9)	Weekly (10)	Bi-weekly (11)	Monthly (12)
$\sqrt{G_{it}}$	0.696 [15.17]	0.715 [14.13]	0.657 [14.40]	0.569 [13.90]								
G_{ij}					1.690 [4.06]	2.134 [3.92]	1.814 [3.85]	1.141 [2.52]				
$G_{ijt}/G_{it}G_{jt}$									0.088 [17.38]	0.085 [10.23]	0.073 [7.70]	0.070 [5.57]
Observations	81,962	81,962	81,962	81,962	3.9M	3.9M	3.9M	3.9M	3.9M	3.9M	3.9M	3.9M
R-squared	0.08	0.08	0.06	0.04	0.04	0.02	0.01	0.01	0.02	0.01	0.00	0.00

Forecasting volatility and comovement at longer horizons.

Read:

- Table 4 shows that co-fragility predicts future covariance and correlation, even after controlling for industry similarity, common owners, size similarity, and book-to-market similarity.
- Table 5 shows that the predictive relation weakens at longer horizons but remains economically and statistically meaningful.

Economic message: Flow-driven ownership structure is especially relevant for short-horizon non-fundamental price pressure, but its effect does not disappear completely at weekly, biweekly, or monthly horizons.

8.5 Figure 4 and Table 6: fragility betas and factor comovement

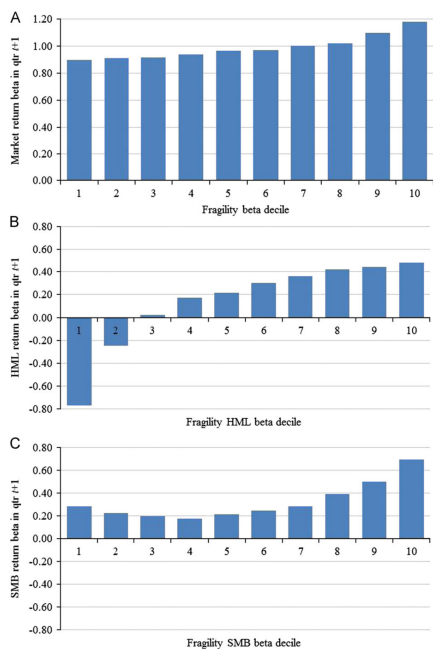


Fig. 4. Fragility betas and return betas. Fragility beta with respect to a portfolio p measures the correlation between flows received by a stock's mutual fund owners and flows into portfolio p . Each quarter from 1989Q4 to 2007Q3, stocks are sorted into deciles of fragility beta with respect to p . Then, for each decile of fragility beta, we first compute the average of the most quarter's return beta with portfolio p . The figure shows the mean future return beta, across dates, by decile of fragility beta. In Panel A, p is the market portfolio. In Panel B, p is the Fama and French (1993) HML portfolio. In Panel C, p is the Fama and French (1993) SMB portfolio. Panel A: Market return beta. Panel B: HML return beta. Panel C: SMB return beta.

Table 6
Co-fragility and three-factor comovement.
Regressions of return betas with respect to portfolio p on fragility betas with respect to the same portfolio p :

$$\beta_{p,t+1}^i = a + bG_{p,t}^i + Z_{p,t}C + u_{p,t+1}$$

In columns 1–2, p is the market portfolio. In columns 3–4, p is the Fama and French (1993) HML portfolio. In columns 5–6, p is the Fama and French (1993) SMB portfolio. Return beta $\beta_{p,t+1}^i$ is the regression coefficient of stock i 's return on the return of portfolio p . Fragility beta $G_{p,t}^i$ is the regression coefficient of flow-driven trades into stock i on flow driven trades into portfolio p . For example, $G_{p,t}^{HML}$ is the regression coefficient of mutual fund flows into stock i on the flows into the portfolio that buys high BE/ME stocks and sells low BE/ME stocks, with similar constructions used for the market and SMB portfolios. The control variables include the fraction of shares outstanding held by mutual funds (MF Share), the number of mutual fund owners, the log of firm size, and the most recently recorded BE/ME ratio. Regressions are estimated quarterly between December 1989 and December 2007; the table reports average coefficients from the 73 quarterly regressions and the associated Fama and Macbeth t -statistics.

	β		β^{HML}		β^{SMB}	
	(1)	(2)	(3)	(4)	(5)	(6)
G^p	0.194 [11.78]	0.125 [7.40]				
G^{HML}			0.482 [21.15]	0.471 [18.52]		
G^{SMB}					0.043 [9.17]	0.042 [9.61]
MF Share		0.880 [3.31]		1.451 [6.28]		-0.551 [-3.36]
N Owners (log)		0.094 [1.39]		-0.106 [-1.85]		-0.016 [-0.30]
Size (log)		-1.010 [-0.02]		133.210 [4.07]		250.160 [9.93]
BE/ME		-0.243 [-8.97]		0.289 [6.99]		0.313 [8.44]
Constant	0.778 [34.62]	0.369 [0.59]	-0.205 [-5.26]	-3.001 [-6.09]	-0.659 [-14.49]	-6.085 [-15.52]
Observations	41,759	30,877	41,155	30,519	41,155	30,519
R-squared	0.06	0.15	0.25	0.29	0.13	0.25
Fama-Macbeth	Yes	Yes	Yes	Yes	Yes	Yes

Fragility betas and return betas.

Co-fragility and three-factor comovement.

Read:

- Figure 4 sorts stocks by fragility beta with respect to market, HML, and SMB portfolios. Stocks whose owners' flows comove with a factor portfolio have higher realized return betas to that factor.
- Table 6 confirms this in regressions. The relation is especially strong for HML: ownership-flow structure explains a large share of value/growth comovement.

Economic message: Some factor comovement may arise from common or correlated investor flows, not only from shared fundamental risk.

8.6 Table 7: arbitrage and the amplification of fragility

Table 7

Liquidity providers and the impact of fragility on volatility: two-stage regressions.

For each stock in each quarter, we first estimate the sensitivity of demand D_t^X of type X on total flow-driven buys from mutual funds D_t^{MF} :

$$D_t^X = \delta_X + \gamma_X D_t^{MF} + v_{tX}.$$

We consider active net buys by mutual funds (X =mutual funds' active purchases), net buys by hedge funds (X =hedge funds), and the sum of these (X =mutual funds active purchases + hedge funds). We estimate these regressions on a rolling basis, so that the estimate of γ_X is based on flows during the past 24 quarters. $|1+\gamma_X|$ serves as an estimate of the extent to which demand of type X accommodates flow-driven trades by mutual funds. When $|1+\gamma_X|$ is large, demand of type X acts as a destabilizing force. The table below reports results from second-stage regressions of one-quarter-ahead volatility of daily returns $\sigma_{R,t+1}$ on fragility $\sqrt{G_{it}}$, $|1+\gamma_X|$, and the interaction of $|1+\gamma_X|$ and fragility $\sqrt{G_{it}}$.

$$\sigma_{R,t+1} = a + b|1+\gamma_X| + c\sqrt{G_{it}} + d|1+\gamma_X|\sqrt{G_{it}} + u_{R,t+1}.$$

Regressions are estimated quarterly between December 1989 and December 2007; the table reports average from the 73 quarterly regressions and the associated Fama and Macbeth t -statistics.

	Dependent variable = $\sigma_{R,t+1}$		
	(1)	(2)	(3)
$ 1+\gamma_1 \sqrt{G}$ (Active MF buys+HF buys)	0.060 [2.95]		
$ 1+\gamma_2 \sqrt{G}$ (HF buys)		0.138 [3.89]	
$ 1+\gamma_3 \sqrt{G}$ (Active MF buys)			0.047 [1.79]
$ 1+\gamma_1 $ (Active MF buys+HF buys)	0.000 [2.71]		
$ 1+\gamma_2 $ (HF buys)		-0.000 [-0.49]	
$ 1+\gamma_3 $ (Active MF buys)			0.001 [3.57]
$\sqrt{G}$	0.438 [10.13]	0.432 [10.61]	0.507 [12.95]
Constant	0.016 [25.79]	0.017 [26.18]	0.016 [28.80]
Mean estimates	Mean($1+\gamma$)=1.033	Mean($1+\gamma$)=1.245	Mean($1+\gamma$)=0.781
Observations	32,046	48,143	43,534
R-squared	0.07	0.09	0.08

Liquidity providers and the impact of fragility on volatility.

Read:

- Table 7 studies whether hedge funds and active mutual fund trades dampen or amplify flow-driven mutual fund trading.
- The interaction of fragility with the estimated amplification term is positive, especially for hedge fund trades.
- This means fragility has a larger volatility effect when other investors trade in the same direction as mutual fund liquidity trades.

Economic message: Arbitrageurs do not automatically stabilize prices. For some stocks, they provide liquidity; for others, their trades reinforce flow pressure and make fragility more damaging.

9 Short summary

Greenwood and Thesmar define stock price fragility as the expected volatility of non-fundamental demand shocks implied by current ownership. Using mutual fund holdings and flows, they show that fragile stocks have higher future volatility. They also extend the concept to co-fragility and fragility betas, showing that correlated investor flows help explain return comovement and factor betas. Finally, they show that arbitrageurs can either dampen or amplify fragility.

10 Conclusion

The paper's central lesson is that ownership matters for risk when demand curves are not perfectly elastic. A stock's risk depends not only on its fundamentals, but also on who owns it and

whether those owners are likely to trade together. Fragility provides a practical way to measure this ownership-based non-fundamental risk.

For asset pricing, the paper suggests that some volatility and comovement are created by investor demand shocks rather than cash-flow news. For market stability, it suggests that concentrated or correlated ownership can make prices vulnerable to liquidity waves. For empirical corporate finance and asset pricing, the paper provides a transferable method: measure the ownership network, infer likely trading pressure, and test whether that pressure predicts prices.